You Don't Need GenServer

This chapter is very important, but it's going to teach you something that you won't use immediately. However, you'll be better prepared to debug your application, run your application in production, and test your application when you understand the material in this chapter.

The reason why you won't use this material immediately is because good Elixir libraries (certainly the ones presented later in this book) use best practices to ensure that they are stable and give you the best possible starting point. Phoenix, Ecto, and Oban will ensure that your application is scalable and error-resilient.

These libraries are designed to not only be appropriately parallel but to also handle errors. They use OTP best practices to help prevent your application from getting in a bad state. You don't know these best practices yet, so your system will be better if you use the libraries.

You'll eventually need GenServers, and you definitely should understand how the process model works from day one. But you may not need to write code that uses them for some time. Instead, lean on the libraries that Elixir experts have written to achieve what you need. At some point, you'll need to write your own GenServer, and you'll be ready to do so.

Let's start this chapter by comparing parallelism and concurrency.

Parallelism vs. Concurrency

It takes a bit of time to fully grasp parallel computing. It's okay if you don't pick it up all at once. But, to give you the best shot, let's start at the ground floor. We'll look at what concurrency and parallelism are and then go into how Ruby and Elixir operate in these areas.

Concurrency Is Not Parallelism

The terms parallelism and concurrency are the source of a lot of confusion. While they seem interchangeable, they refer to distinctly different, but related, topics.

Concurrency is the ability to *coordinate* multiple tasks at the same time. Imagine that there's a single CPU core on a server. Concurrency allows that single CPU core to weave together and execute tasks that are necessary to serve requests. From an outsider's perspective, it would appear that requests are handled at the same time.